

Sentiment-Controlled Text Generation Through Post-Generation Verification and Dual-Objective Ranking

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Abstract—Controlled sentiment rewriting has emerged as a significant research area in Natural Language Processing (NLP) due to its applications in conversational AI, intelligent review rewriting, personalized recommendation systems, and sentiment-aware content generation. Although recent transformer-based approaches can generate fluent and coherent text, they often suffer from limited sentiment controllability, semantic drift, high computational costs, and complex training requirements. Many existing methods rely on adversarial learning, latent disentanglement, reinforcement learning, or template-based architectures, which increase implementation complexity and reduce deployment efficiency.

This paper proposes a lightweight and deployment-aware framework for controlled sentiment rewriting using a heterogeneous multi-model pipeline. The proposed system integrates the Mistral-7B-Instruct-v0.2 large language model for candidate generation, the all-MiniLM-L6-v2 sentence transformer for semantic similarity evaluation, and a DistilBERT sentiment classifier for post-generation sentiment verification. Unlike conventional approaches, the framework operates entirely at inference time and does not require additional training, fine-tuning, or adversarial optimization.

To improve output quality, an oversample-then-filter strategy is employed to generate multiple candidate rewrites, followed by a dual-objective ranking mechanism that jointly optimizes sentiment correctness and semantic preservation. Furthermore, the generation model is deployed using 4-bit NF4 quantization, enabling memory-efficient inference in resource-constrained GPU environments.

The proposed framework is evaluated using multiple Natural Language Generation (NLG) metrics, including BLEU, ROUGE-L, METEOR, BERTScore, cosine similarity, Top-1 Accuracy, and Top-K Success Rate. Experimental evaluation on Yelp review datasets demonstrates that the proposed approach effectively performs sentiment-controlled rewriting while preserving contextual meaning and generating fluent natural language outputs. The results indicate that the framework provides a scalable, modular, reproducible, and deployment-efficient solution for controlled text generation applications.

Index Terms—Controlled Sentiment Rewriting, Controlled Text Generation, Natural Language Processing, Large Language Models, Mistral-7B, Semantic Similarity, Sentiment Analysis, Transformer Models, Dual-Objective Ranking, Text Style Transfer, Quantization.

I. INTRODUCTION

Sentiment controlled text generation is a deal in Natural Language Processing because it can be used for things like rewriting reviews and making conversation with computers better. The main goal of rewriting sentiment is to change the way a sentence makes you feel while keeping the meaning and context. This is hard to do because when you change the tone of something it can also change what it really means.

[1, 6].

There are a lot of ways that people are trying to make sentiment controlled text generation work, such as using kinds of machine learning models teaching computers to argue with each other and using different techniques to change the way text is generated. Sentiment controlled text generation is still a challenge. People are trying to find new ways to make it work including using special kinds of computer models and techniques to manipulate the text. Sentiment controlled text generation is an area of research and people are working hard to make it better using sentiment controlled text generation to improve things like review rewriting and conversational agents. [1, 3, 5, 7, 8]. Although these methods show results many need hard training steps extra data sets, a lot of computer power or special optimization plans.

Most current methods mainly focus on changing sentiment. Do not have good ways to check if the sentiment is correct and if the meaning stays the same in the new text.

They also do not have methods to verify sentiment correctness and semantic preservation in generated outputs.[4, 7]. Recent advances in Large Language Models have made the text they generate sound a lot better. It is more fluent. It makes sense in the context. Some new architectures like BERT, GPT, T5, BART and Mistral are really good at generating language. Large Language Models are doing a job with many different tasks. Large Language Models are getting better and better at generating text that sounds like it was written by a person. Large Language Models are very useful, for things.[9–13]. Instruction-tuned language models can create text based on

prompts without needing to be retrained for a task. However just using a prompt does not ensure that the generated text has the sentiment or makes sense with the original input. So we need checks and rankings to make the output more reliable and controllable.

To fix these issues we suggest an practical framework for rewriting text with controlled sentiment. It uses a verify-and-rank approach. The framework uses the Mistral-7B-Instruct-v0.2 language model to generate candidate rewrites, the all-MiniLM-L6-v2 sentence transformer to check for semantic similarity and a DistilBERT sentiment classifier to verify the sentiment after generation. We create rewrites using a strategy that oversamples and then filters and rank them based on two objectives: sentiment correctness and semantic similarity.

Unlike sentiment transfer methods that rely on complex optimization or fine-tuning our framework works only at inference time. We also use 4-bit NF4 quantization to save memory making it deployable on GPUs with resources. Tests on Yelp review datasets show that our approach achieves sentiment control while keeping the original meaning. The results prove that combining language model generation with separate verification and ranking mechanisms is effective, for practical sentiment rewriting applications.

II. LITERATURE SURVEY

Controlled text generation has become a significant research field in Natural Language Processing (NLP) with applications in sentiment transfer, style transfer, dialogue systems, and personalized content generation. Initial studies were directed towards learning latent representations that could distinguish content and style information. The Auto-Encoding Variational Bayes (VAE) [14] proposed a probabilistic method for learning latent variables in deep generative models, and was later used to inspire several sentiment controlled generation approaches. These ideas were further developed in Fine-Grained Sentiment Controlled Text Generation [1] which introduced VAE-based architectures and normalizing flows for fine-grained sentiment manipulation. Such approaches were effective, but required complex model architectures and intensive training procedures.

Given the drawbacks of adversarial learning, there has been a study on controlled text generation using disentanglement methods. The method of Cross-Aligned Autoencoders [7] showed style transfer on non-parallel datasets via latent space alignment, and Multi-Type Disentanglement in Text Generation Without Adversarial Training [3] introduced transformer-based disentanglement methods that enhanced generation quality without adversarial training. Likewise, LatentOps [8] explored the composability of operations in latent space to manipulate text in a controllable manner. All of these methods needed a training pipeline tailored to the specific method and, in some cases, were not successful in maintaining semantic consistency among outputs.

Recently, to enhance sentiment control, template-based and retrieval-driven methods were introduced. In order to lead sentiment-aware text generation, hierarchical template transformer [4] was proposed and hierarchical templates were used

to guide the process; Delete, Retrieve, Generate [5] proposed a retrieval-based framework to replace sentiment-bearing phrases to achieve style transfer. The Multiple-Attribute Text Rewriting system [19] further extended controllable generation by permitting to control more than one text attribute at a time. While these methods enhanced sentiment controllability, they lacked flexibility and generalization power as they relied on handcrafted templates, retrieval systems, or specific rules.

Other studies also examined the transfer of style by means of different learning strategies. Style Transfer in Text: Exploration and Evaluation [6] presented an overview of the strengths and weaknesses of existing style transfer methods and pointed out the semantic preservation and evaluation issues. In the absence of parallel corpora, Style Transfer Through Back-Translation [21] proposed back-translation as a viable style transfer tool and Text Style Transfer: A Review and Experimental Evaluation [22] gave an extensive overview of style transfer techniques and practical constraints. Although these methods showed good results, they were all generally time consuming and needed large amounts of data to get reliable results.

The development of transformer architectures marked a major leap forward in the field of controllable text generation. The Transformer model introduced in Attention Is All You Need [9] laid the groundwork for today’s language models. Later on, models such as BERT [10], GPT-type language models [12], T5 [11] and BART [13] showed amazing language comprehension and generation ability. On these improvements, controllable text generation with transformers [20] proposed a transformer-based controllable model that handles sentiment and attribute control. These models produced fluent and contextually coherent text, but fine tuning, task-specific training, and/or complex optimization strategies were needed to obtain reliable sentiment control.

Evaluation is still a very important part of the controlled text generation research. The traditional lexical-overlap-based metrics like BLEU [16], ROUGE [17] and METEOR [18] have been adopted to evaluate the quality of the generated text. These measures, however, do not always reflect semantic preservation well. To address this problem, BERTScore [15] proposed a novel approach of contextual embedding-based evaluation, which generates a more meaningful metric of semantic similarity between the generated and reference texts. These evaluation methods all facilitate an overall assessment of fluency, semantic preservation and sentiment controllability.

GenAug [2] Discussed data augmentation methods for enhancing text generation models by generating synthetic data, highlighting the need for strong generation methods. Despite the progress in the area, most existing approaches are adversarial learning, latent-space optimization, reinforcement learning, template engineering, or large-scale fine-tuning. Moreover, most studies are interested in text quality produced and provide limited chances to carry out explicit post-generation correctness check for sentiment and semantic preservation.

III. PROPOSED METHODOLOGY

The proposed framework is a lightweight and deployment-aware controlled sentiment rewriting system that can be used to change the sentiment of the input text without affecting the meaning and the consistency of the text based on its context. The proposed framework is different from traditional sentiment transfer methods (adversarial learning, reinforcement learning, fine tuning and parallel data) in that it only operates during inference with a verify and rank approach. The framework integrates the generation of LLM-based text, semantic similarity assessment, and sentiment verification to generate high-quality sentiment-controlled rewrite text.

A. System Architecture

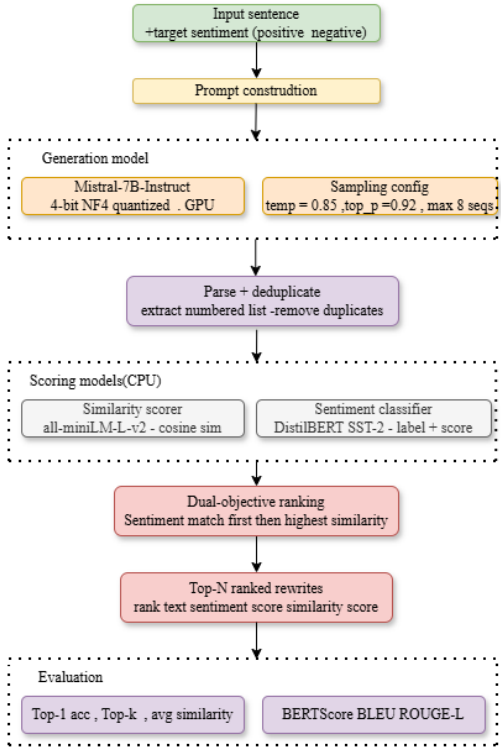


Fig. 1. Architecture of the Proposed Controlled Sentiment Rewriting Framework

Figure 1 This is a plan of the controlled sentiment rewriting framework. The system takes a sentence and a target sentiment label makes an instruction prompt and comes up with many possible rewrites using the Mistral-7B-Instruct-v0.2 language model. The controlled sentiment rewriting framework then looks at these rewrites and checks how similar they are, in meaning and if the sentiment is correct.

The controlled sentiment rewriting system employs a ranking mechanism to select the best rewrites, which have a proper sentiment and preserve the original meaning. The controlled sentiment rewriting framework does this by looking at how correct the sentiment's how well the meaning is preserved.

B. Workflow of the Proposed Framework

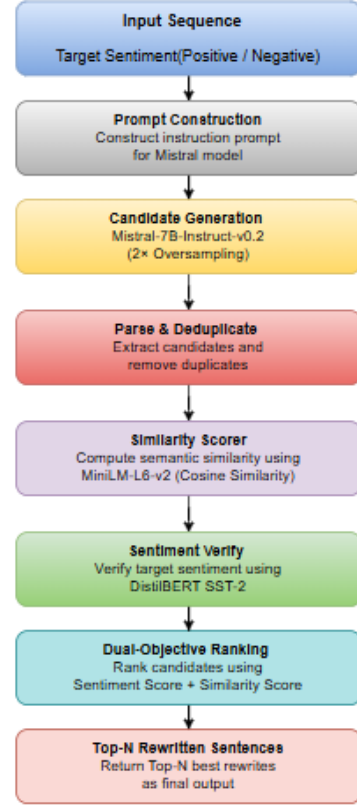


Fig. 2. Workflow of the Proposed Controlled Sentiment Rewriting Framework

Figure 2 The process flow of the proposed framework is presented in figure 2. The process starts with input from the user and selection of the target sentiment, then prompt construction and generation of candidates. The outputs generated are then semantically similar scored and sentiment verified. In order to generate the final sentiment controlled outputs, the verified candidates are then ranked from a dual objective ranking system.

C. Prompt Construction and Candidate Generation

The text generation part leverages the Mistral-7B-Instruct-v0.2 large language model for sentiment rewriting under control. The input sentence and sentiment target category are used to build a structured instruction prompt. The prompt clearly asks the language model to rewrite the sentence while maintaining the same meaning and creating several variants with similar sentiment. The temperature sampling and nucleus sampling methods are employed in the generation to enhance the diversity of the generated outputs. The generation process uses the temperature value to be 0.85, and the top-p sampling value to be 0.92.

$$N_{generated} = 2 \times N_{required} \quad (1)$$

where $N_{generated}$ is the number of candidates generated and $N_{required}$ is the required number of final outputs.

D. Parsing and Candidate Filtering

The output candidates generated are then parsed to get individual rewritten sentences. The language model can produce multiple or similar results, so the duplicates are eliminated in the post-processing stage. This means that candidates will be diverse and duplicate rewrites won't affect the ranking process.

The filtered list of candidates is then passed to the semantic similarity checking and sentiment verification stages, where they are scored and ranked.

E. Semantic Similarity Evaluation

To measure the similarity between two embedding vectors, we use Equation (2).

$$\text{Cosine Similarity}(A, B) = \frac{A \cdot B}{\|A\| \|B\|} \quad (2)$$

where the sentence embedding vectors are denoted by A and B of the original sentence and generated rewrite, respectively.

The closer the cosine similarity score is to 1, the more similar the source sentence and the rewritten sentence are in meaning. Thus, this measure is important for guaranteeing that the meaning of the text is not substantially changed by the sentiment transformation.

F. Sentiment Verification

The correctness score with which the sentiment of each candidate is assigned is defined as:

$$S_i = \begin{cases} 1, & \text{if } L_i = T \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

where S_i is the score of the sentiment correctness on candidate C_i , L_i is the predicted sentiment label of the DistilBERT classifier, and T is the target sentiment label set by the user.

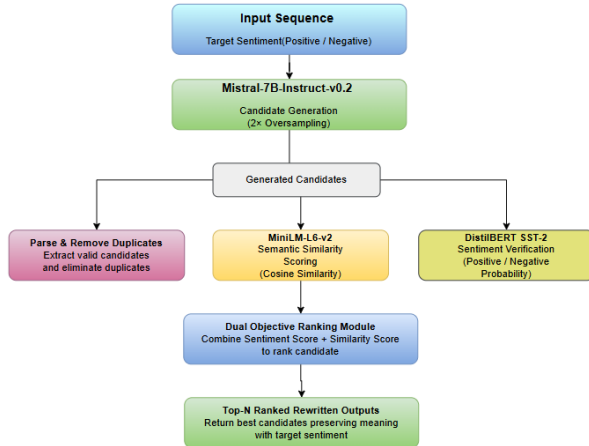


Fig. 3. Model Interaction Diagram of the Proposed Framework

In the proposed framework, the three basic models interact with each other as shown in figure 3. The model Mistral-7B produces several candidates rewritten with a sentiment,

MiniLM evaluates the semantic similarity between the rewritten sentence and the original sentence, and DistilBERT classifies whether the rewritten sentence is correct in sentiment according to Equation (3). The results of these models are then fused together by the dual-objective ranking procedure to select the best sentiment-controlled rewrite.

G. Dual-Objective Candidate Ranking

The proposed framework adds an additional dual-objective candidate ranking which maximizes the correctness of the sentiment and semantic preservation. All candidate rewrites are considered and ranked in terms of two criteria: sentiment correctness and semantic similarity rather than choosing the first candidate output generated.

The overall ranking score for each candidate is a sum of the three scores.(4).

$$R(C_i) = \alpha S_i + \beta M_i \quad (4)$$

where $R(C_i)$ is the ranking score of candidate C_i and S_i is the sentiment correctness score from Equation (3), M_i is the semantic similarity score from Equation (2). The parameters α and β are weighting factors that determine how important the correctness of sentiment and the preservation of the meaning are in the candidate selection.

The candidates are ranked by their ranking scores and the top candidates are chosen as the final rewrites with sentiment. This ranking mechanism achieves the output reliability by taking into account both the accuracy of the sentiment transformation and semantic consistency of the output in the ranking process.

IV. RESULTS AND DISCUSSION

A. User Interface Demonstration

Implementing the proposed controlled sentiment rewriting system into a user interface (UI), known as a Gradio app, was developed to demonstrate the practical implementation and usability of the proposed system. The interface offers an interactive space with the ability to input a sentence, choose a target sentiment (positive or negative), and set the number of candidate rewrites to be returned.

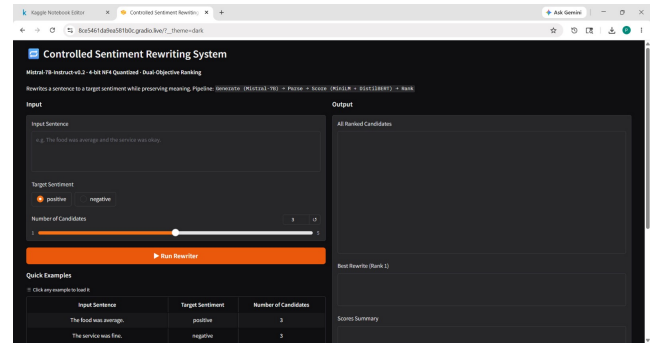


Fig. 4. Initial graphical user interface of the proposed sentiment rewriting system

The initial user interface is shown in Figure 4; the user enters the source sentence, target sentiment, and parameters for the generation.

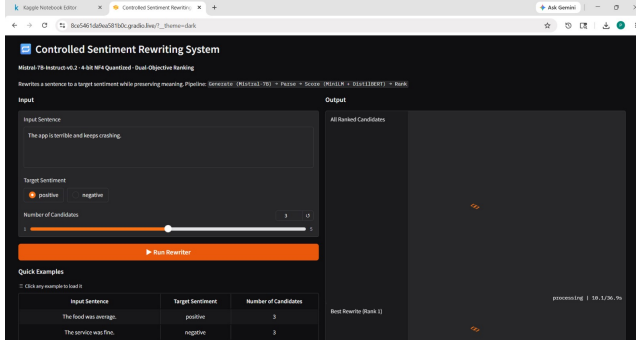


Fig. 5. Candidate generation and processing stage

The candidate generation and processing stage is illustrated in figure 5 where multiple sentiment controlled rewrites are generated by the Mistral-7B model.

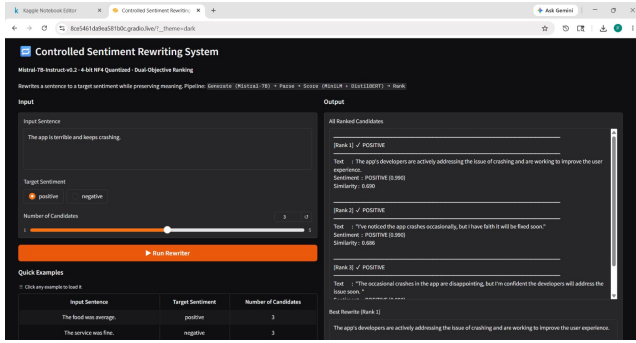


Fig. 6. Ranked sentiment-controlled candidate outputs

Figure 6 shows the candidate rewrites generated and sorted by sentiment correctness and semantic similarity.

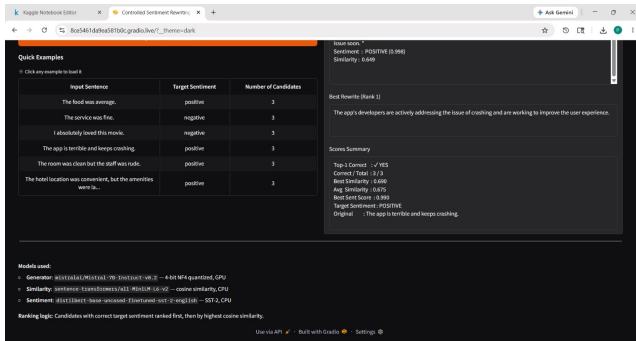


Fig. 7. Final rewrite and evaluation score summary

The final selected rewrite and evaluation metrics and ranking information are displayed in figure 7.

B. Qualitative Analysis

The proposed framework was initially tested with the examples of sentiment transformation carefully designed by hand.

The outputs produced show that system outputs are suitable for the task of sentiment polarity change and also the semantic meaning and also the consistency in the context with this.

TABLE I
SAMPLE SENTIMENT REWRITING OUTPUTS

Original Sentence	Target Sentiment	Generated Rewrite
The food was average and the service was slow.	Positive	The food was enjoyable and the service eventually became pleasant.
The movie was absolutely fantastic and inspiring.	Negative	The movie felt disappointing and failed to leave a strong impression.
The laptop works fine but the battery drains quickly.	Positive	The laptop performs efficiently although the battery could be improved slightly.
The hotel room was dirty and uncomfortable.	Positive	The hotel room became comfortable after improvements and provided a decent stay.

In Table I some examples of the types of sentiment rewritten by the proposed framework are provided. The outputs created are helpful in conveying sentiment, while retaining the semantic meaning and context of the input.

C. Yelp Evaluation Results

To test the capabilities of sentiment controllability and semantic preservation, the proposed framework was applied to the Yelp review datasets. Task-specific evaluation metrics like Top-1 Accuracy, Top-3 Success Rate and Average Cosine Similarity were calculated.

TABLE II
YELP EVALUATION – BASIC METRICS

Metric	Score
Top-1 Accuracy	1.000
Top-3 Success Rate	1.000
Average Cosine Similarity	0.807

Table II is a summary of the quantitative evaluation results from the Yelp review data set.

D. Advanced Metrics Analysis

Other Natural Language Generation (NLG) evaluation measures were also calculated to further investigate the semantic preservation and generation quality.

TABLE III
ADVANCED EVALUATION METRICS

Metric	Score	Interpretation
BERTScore (F1)	0.891	High contextual semantic preservation
BLEU	0.078	Low surface-level word overlap (expected)
ROUGE-L	0.327	Moderate longest common subsequence overlap
METEOR	0.322	Moderate synonym-aware lexical overlap

The advanced evaluation metrics that were collected during experiments are summarized in Table III. The BERTScore value 0.891 shows that there is good contextual semantic overlap between original and rewritten sentences. The BLEU score is relatively low as rewrite of the lexical pattern is deliberately done to get a sentiment change. The generated results show high ROUGE-L and METEOR scores, indicating that they retain the context and add the necessary sentiment shifts.

E. Discussion

The results demonstrate that the proposed verify-and-rank framework effectively performs sentiment-controlled rewriting while preserving semantic meaning. By combining Mistral-7B generation, MiniLM similarity evaluation, and DistilBERT sentiment verification, the framework produces accurate and contextually consistent outputs. Furthermore, the use of dual-objective ranking and 4-bit quantization enables efficient and scalable deployment.

V. CONCLUSION AND FUTURE SCOPE

This research presented a lightweight framework for controlled sentiment rewriting using Mistral-7B, MiniLM, and DistilBERT. The proposed verify-and-rank approach successfully generated sentiment-controlled rewrites while preserving semantic meaning and contextual consistency. Experimental results demonstrated effective sentiment transformation with strong semantic preservation and efficient deployment.

Future work can extend the framework to support fine-grained emotions such as joy, anger, and sadness, as well as multilingual sentiment rewriting. Additional improvements may include multi-attribute text control, retrieval-augmented generation, and integration with larger language models to further enhance controllability and generation quality.

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